

PURBANCHAL UNIVERSITY



DEPARTMENT OF COMPUTER ENGINEERING

KHWOPA ENGINEERING COLLEGE

LIBALI-08, BHAKTAPUR

A project report on

LITERATURE EFFICIENT ARCHITECTURE FOLIO [LEAF]

A report submitted for the partial fulfillment of requirements for the degree of Bachelor of
Engineering in Computer Engineering (Third Semester)

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UNDER THE SUPERVISION OF:

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November 24, 2023

APPROVAL LETTER

I hereby confirm that I have reviewed and recommended for approval to the Institute of Engineering a project report titled "LITERATURE EFFICIENT ARCHITECTURE FOLIO" (LEAF). This report has been submitted by Jayadev Tripathi, Sahana Shrestha, Sanjib Kasti and Upendra Shahi in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering (Third semester).

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Signature

Date

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Heartfelt thanks are due to our esteemed teachers, whose guidance and encouragement were instrumental throughout this journey. Additionally, our gratitude extends to our supportive friends whose camaraderie added a valuable dimension to our project experience. We are also grateful to all those, both directly and indirectly, who contributed their expertise, time, and encouragement, making this project a reality.

Thank you.

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ABSTRACT

Our project LEAF (Literature Efficient Architecture Folio) is an architecture for the manipulation of literature-based contents and creations. It is an efficient platform for all literature enthusiasts. The system is based on the localized server (i.e. data are stored only in the local system or user's system). Supporting multiple users to engage in a system and have their interaction, the system provides secured sessions. It is totally concerned with promoting enthusiasm in literature and helping in organizing users' creations enabling them to create their project and collaborate by sharing the Lits among the members who can access that through their own credentials in the same system.

This console-based application will also allow the user to view their analytics and engagements, read the loaded contents, and bookmark the preferred and impressive ones. Tying with the fluent, organized, and user-friendly experience; Project LEAF will be part of taking your literature experience to new heights with some fascinating graphical interfaces.

Keywords: - Analytics, LEAF, Literature, Articles, LeafHandle, Lits

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CHAPTER 1

INTRODUCTION

1.1 Background

In today's digital world, every field has been randomly digitized. To cope with the efficient digitization in the field of literature LEAF has been developed with prime motive to enrich the user experience with the efficient architecture and user-friendly interface. This project aims to ease users with exciting features and promotes their enthusiasm in literature. The system allows multiple user to function in. User can choose to login/ signUp and enter into the system.

Inside the Lits section, the user can communicate with each other through their shared files. The users can even give and view the system feedback. The system also allows proper account management with functionalities to reset password, pin, handle and profile image.

This system allows to keep notes, create exciting literatures and share among the registered users. It even allows to view your analytics and manage the file access. This project is all about some sophisticated graphical interfaces for a literature repository using C++.

1.2 Problem Statement

In this digital era, everyone desires to organize their notes in the regular basis. They want to experience their literature experience in a single platform having multiple functionalities and simple and efficient interface. They want to keep records, add teams, have conversation, manage tags, have search functionality and track progress too. And this is mandatory if one wants to move ahead in any respective field.

1.3 Objective

The main objective of project LEAF is to develop a user friendly and multi-functionality literature repository.

1.4 Significance And Scope

This project LEAF will be outstanding platform to explore the literature field. This will be useful for one seeking for the tool to enable them to record their creation and gather as a project. This will entirely give outstanding experience of the literature field and helps the individuals to explore their creativity, record it in organized manner, and view the progress track. This also enables to upload ones favorite contents and read it during the leisure.

CHAPTER 2

LITERATURE REVIEW

In this era of advanced technologies, the world is caught up in a single computer or digital device. Every moment and all human activities are associated with the digital platform. Aside, literature is the magic that can overcome sorrows, pain, and loneliness and motivates in every failure and further encourage to achieve the goal. In the digital era, the development of applications provides an amazing platform for organizing and creating literature.

The application like memo in the early keypad phones to the evolved application like: Google Keep, Evernote, Microsoft OneNote, To do List, and soon clarifies the importance of the habit of note takings. On the other side, Evernote, Instapaper, Pocket, Flipboard, Read wise, etc. applications are popular for accessing and reading the literature contents.

As a user, every individual wants a well-organized application that not only allows one to create notes and general articles or just provides written articles to look over, but they also want wide range of features that enhance the overall literature experience. Google Keep is concerned only with keeping, editing, and sharing notes while Evernote facilitates reading articles more. However, they lack allowing user to create their typical literature projects, classifying the literature genre, set the nickname, track their analytics, add the collaborator securely, manage tags and search for the prominent literature figures in the same platform. As the one to address these issues and promote the journey through the magical world of literature, the project “LEAF” has been developed.

In the exact similar concept in C++, none such projects are build earlier. However we found a social media in simple cpp by “kostathanos” in github. However that project lacks the proper error handling, validation and graphical interfaces.

CHAPTER 3

PROJECT MANAGEMENT

3.1 Project members:

All the required efforts for this project were done by all of the four members of a team:

- Jayadev Tripathi [780314]
- Sahana Shrestha [780336]
- Sanjib Kasti [780339]
- Upendra Shahi [780347]

3.2 Work Breakdown Structure:

S.N.	Month Job description	1 st month	2 nd month	3 rd month	4 th month	5 th month	6 th month
1	Problem Identification						
2	Analysis						
3	Design						
4	Coding						
5	Implementing and testing						
6	Documentation						

Fig 3.1: Work Breakdown structure

CHAPTER 4

METHODOLOGY

4.1 Algorithm

Step 1: Start

Step 2: Choose either to login, SignUp or Help.

Step 3:

- (i) If choice is login:
 - a) Take user Input for LeafHandle and password.
 - b) Perform password validation for entered credentials.
 - c) If authenticated then goto homepage else, go back to first page.
- (ii) If choice is SignUp and if valid and correct display next option as:
 - a) Take userInput for the required fields and perform validation.
 - b) After successful registration redirect to first page.
- (iii) If choice is Help:
 - a) Display Help file.

Step 4: Display Home Page

- a) Leaf
 - a. Notes
 - i. Allows to create the note. And discard if it is empty.
 - b. Templates
 - i. Allows to create some document from built in templates.
 - c. Lits
 - i. My Lits
 - 1. Display all the saved Lits and allows either to open, delete, share or edit.
 - ii. Create
 - 1. Allows to create a new Lit.
 - iii. Shared
 - 1. Display all the shared Lits and Manage access if you are the owner.
 - d. Tags
 - i. Display all the created tags and files inside it.
 - e. Genre
 - i. Display all the created genre and files inside it.
- b) Read
 - a. Literature
 - i. Allows to read the preloaded contents sequentially and can add the preferred to favorite.
 - b. Authors
 - i. Allows to randomly view Authors and search for the author.
 - c. Composed
 - i. Sequentially displays all the composed Lits.
 - d. Favourites

- i. Sequentially displays all the favourites added.
- c) My Account
 - a. My Notes
 - i. Displays all the created notes.
 - b. My Templates
 - i. Displays all the created Templates.
 - c. My Notifications
 - i. Shows the recent activity from shared section.
 - d. My Analytics
 - i. Shows the analytical graph.
 - e. Settings
 - i. Change Username
 - ii. Change Password
 - iii. Change Profile Image
 - iv. Change PIN
 - v. Logout
- d) Feedback
 - a. Allows user to give their review and view other users review on system.
- e) About Us
 - a. Display About our team.

4.2 Tools and platform

Tools: CodeBlocks

Platforms: Windows

CHAPTER 5

RESULT AND DISCUSSIONS

5.1 Outcomes

Some Screenshots of the program are showed here.

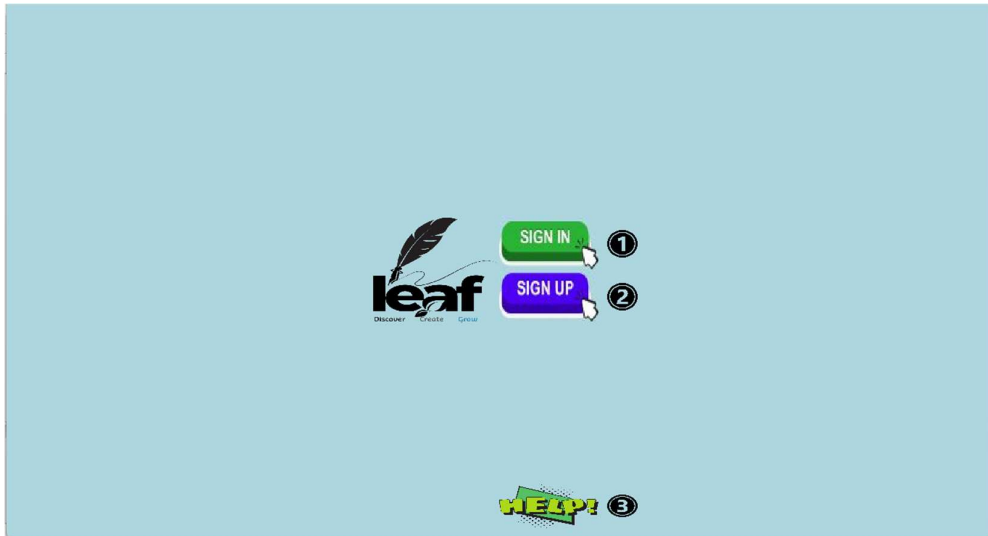


Figure 5.1.1 First Page (Login SignUP)

This page allows to choose between Login/ SignUp and also gives instructions for users.

A screenshot of a web application's SignUp page. The background is a solid light blue. At the top, there is a yellow banner with the 'leaf' logo and the text 'LITERATURE EFFICIENT architecture folio' and 'Discover Create Grow'. Below the banner, there are six input fields, each with a label in a blue box: 'Full Name :', 'LeafHandle :', 'Password :', 'Confirmation :', 'Profile Image :', and 'Reset Code :'. Each label is followed by a light purple input field.

Figure 5.1.2 SignUp Page

This page allows to register new user and redirect to first page.

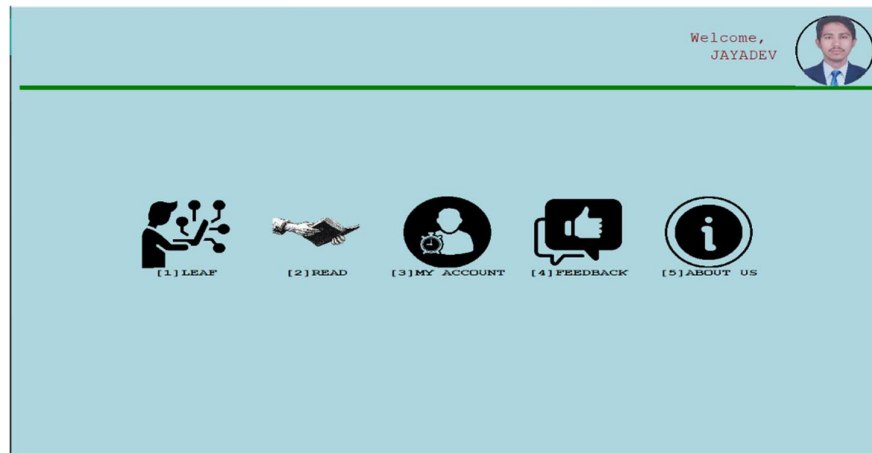


Figure 5.1.3 LoggedIn Page

This is the logged in interface for the logged IN user after credentials are verified.



Figure 5.1.4 Leaf Section

Inside this section, user can create notes and templates. They can even create Lits and share among the existing users. They can view created tags and genres too.

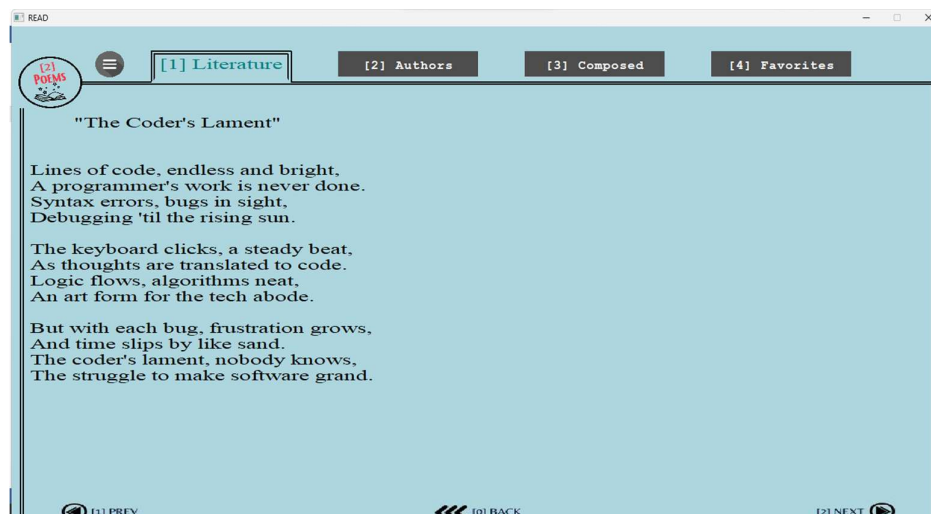


Figure 5.1.5 Read Section

In this section, users can read the preloaded contents. They can also read their creates lits inside this section. They can add/remove the contents to favorite and view inside it.

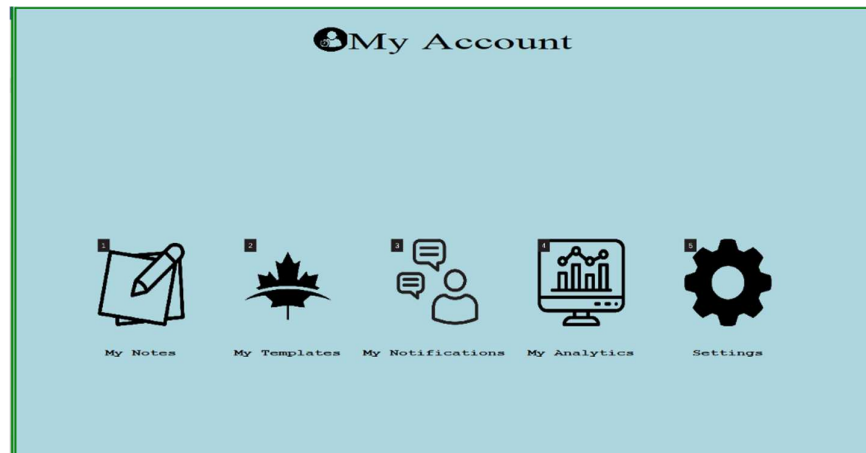


Figure 5.1.6 My Account Section

User Created Notes, templates, notifications received, graphical analytics and settings are inside this section.



Figure 5.1.7 Settings Section

User Can change their login credentials and logout from the program through this section.

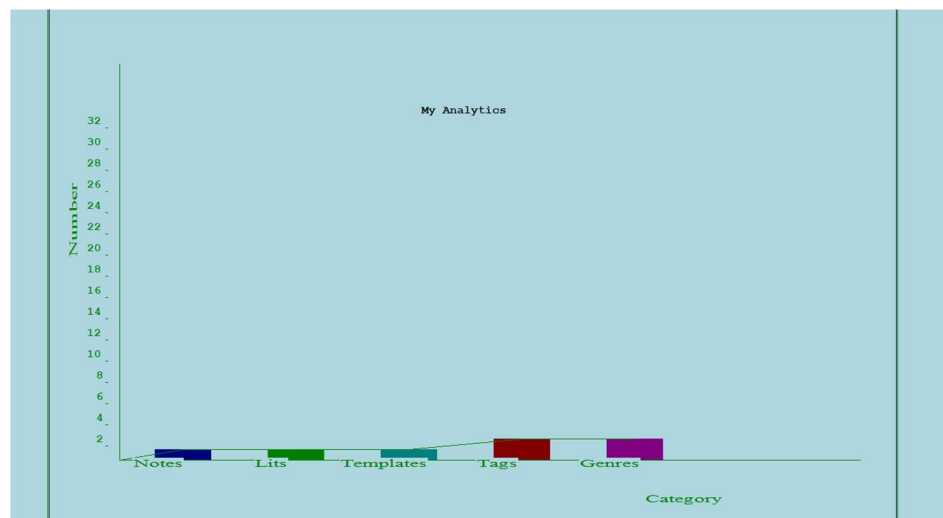


Figure 5.1.8 Analytics Section

In this section, User can view their analytical graphs of the notes/templates created and genre and tags.

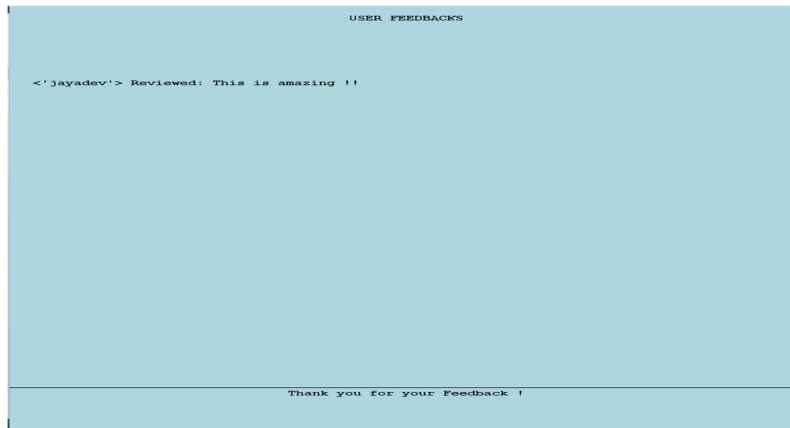


Figure 5.1.9 Feedback Section

In this section user can give their valuable feedback to the system and also view what other users reviewed about the system.

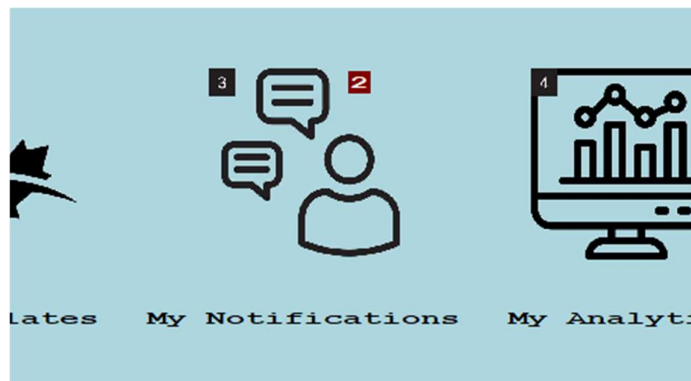


Figure 5.1.10 Notification Received Representation

The number at the right corner represents the unseen notifications.

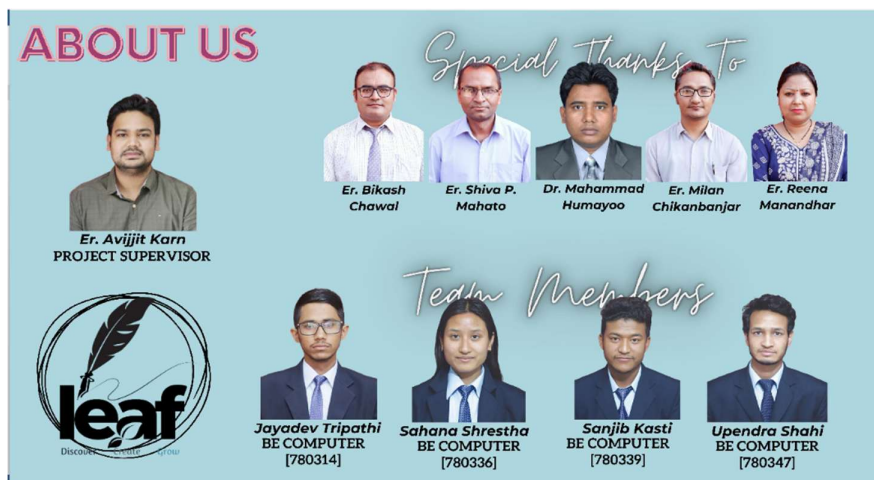


Figure 5.1.11 About Us Section

This section shows our supervising and guiding faculties and our team members.

Thanks to all !!

5.2 Conclusion

To some extent the project objective, to create a literature repository with some graphical representation was achieved with huge support from the supervisor and all the faculties. The entire system was developed in the windows platform using 'graphics.h' library for graphics manipulation and C++ as the preferred programming Language.

The system succeed to provide effective friendly interface and proper user management. Targeted objective to allow the multiple users to interact with the shared files was also achieved. Overall, the intended functionalities and objective was met.

5.3 Limitation and Future Recommendations

The system however has several limitations. To overcome those we suggest following future enhancements over this system.

- i. Scrolling window for displaying larger documents.
- ii. Mouse clicks for more efficient user interface.
- iii. Consistent color use.
- iv. Use of enhanced graphics libraries like “OpenGL”, “DirectX”, SDL and soon for smoother graphics and more sophisticated interface.
- v. Some more functionalities and online access.

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